

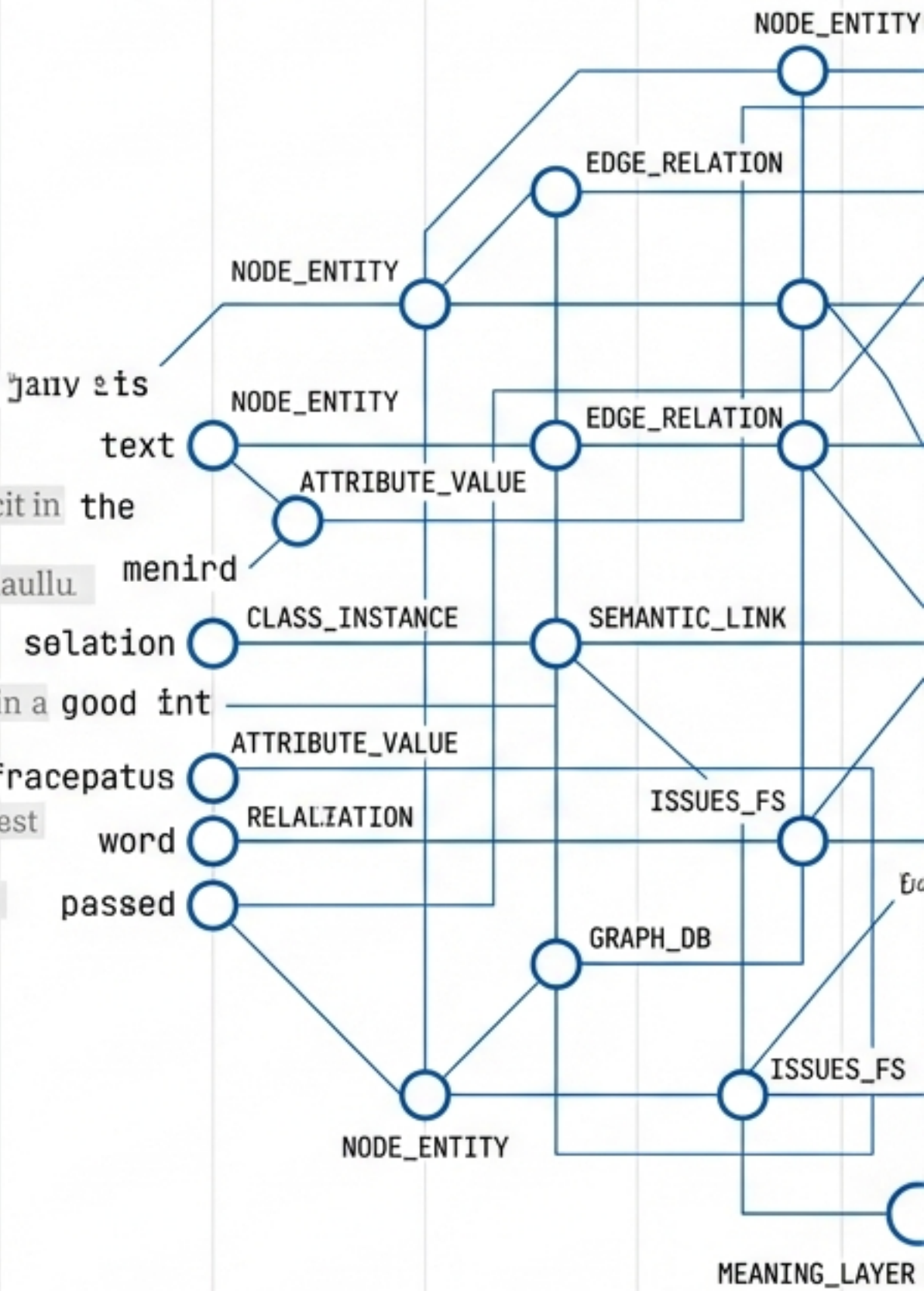
Semantic Text Architecture: Text as Graph

Moving from Opaque Strings to Transparent Meaning within the Issues-FS Ecosystem.

Version: v1.0
Status: Draft
Date: 05 February 2026

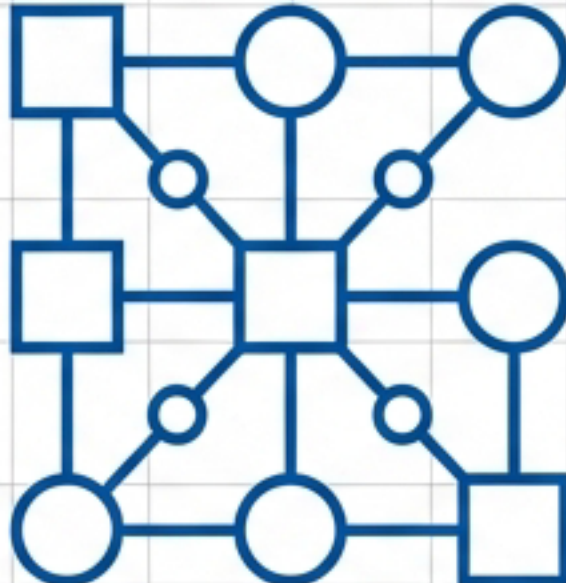
Deconstruction

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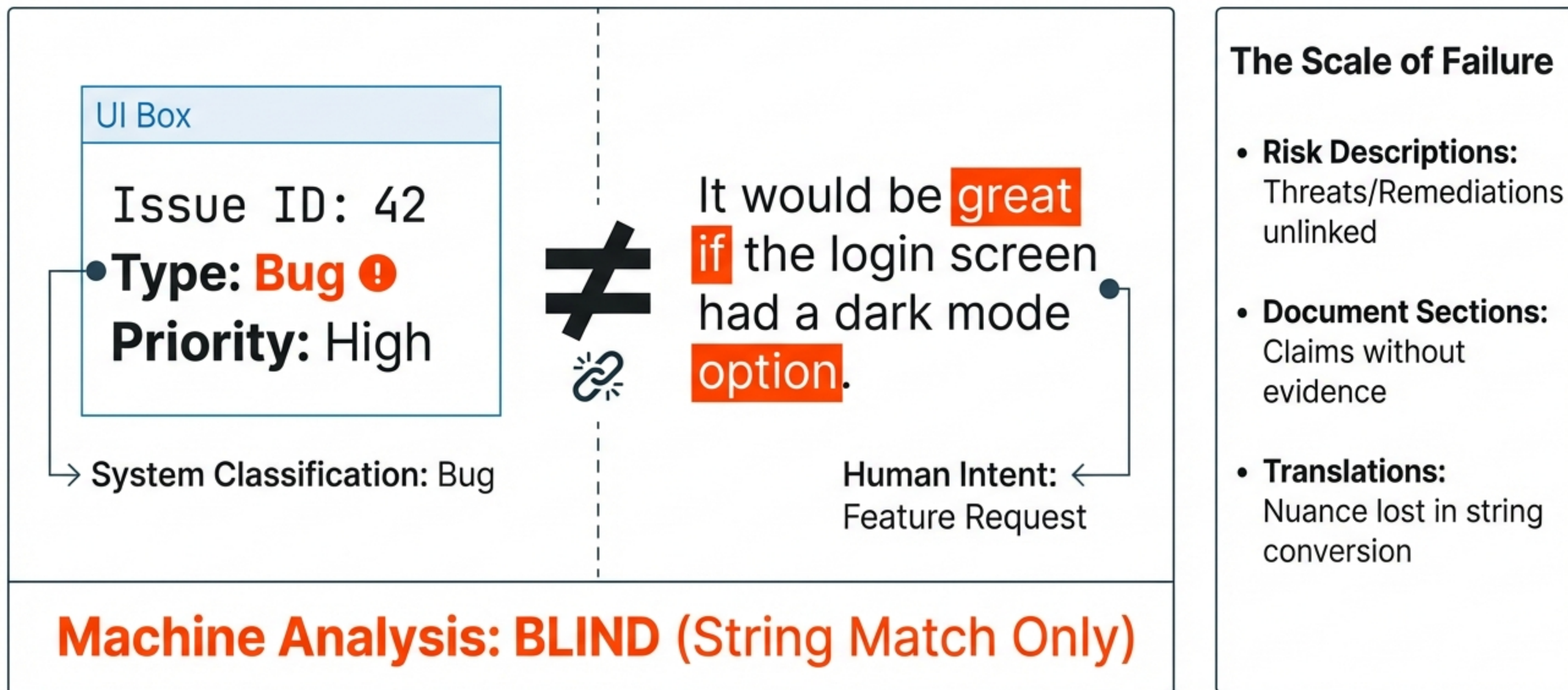


The 'Meaning Through Connectivity' Principle Must Apply to Content

We are applying strict graph principles to text to achieve the same rigour we have for code.

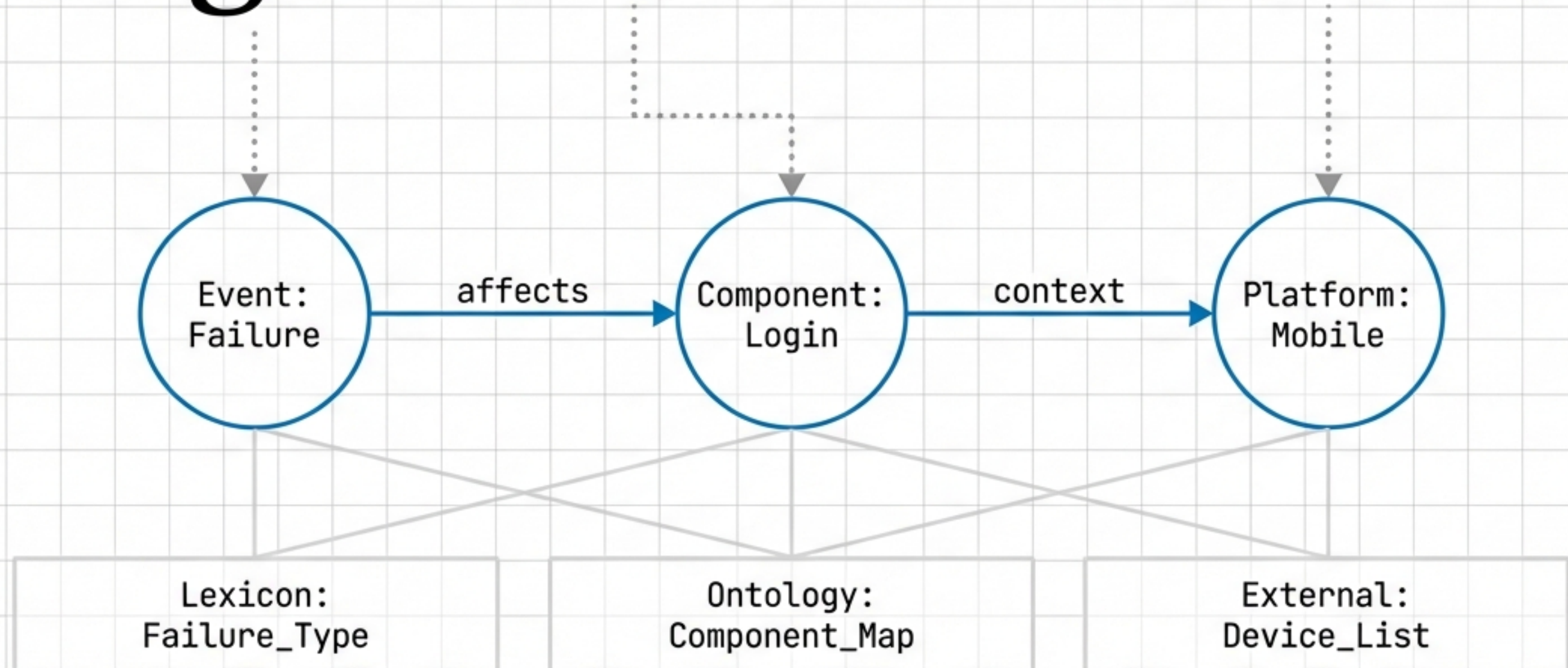
Current State		Future State	
	Text as String ✗ Terminal Leaf Node ✗ Opaque Content ✗ Isolated Data		Text as Graph ✓ Transparent Network ✓ Traversable Meaning ✓ Semantic Validation
Key Capabilities			
			
Semantic Validation	Change Detection	Hallucination Detection	Testable Content

The Blind Spot: Why 'Text as String' Fails



The Solution: Every Unit of Text is a Graph

Login failed on mobile.

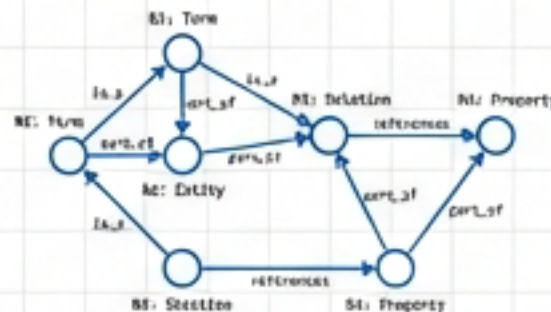


The Fractal Document Model: Graphs of Graphs

Level 1: Document Graph (Summary/Projection)

Level 2: Section Graph

Level 3: Paragraph Graph



Architectural Benefits

- 1. Independent Storage (Zip/DB)
- 2. Independent Versioning
- 3. Composability

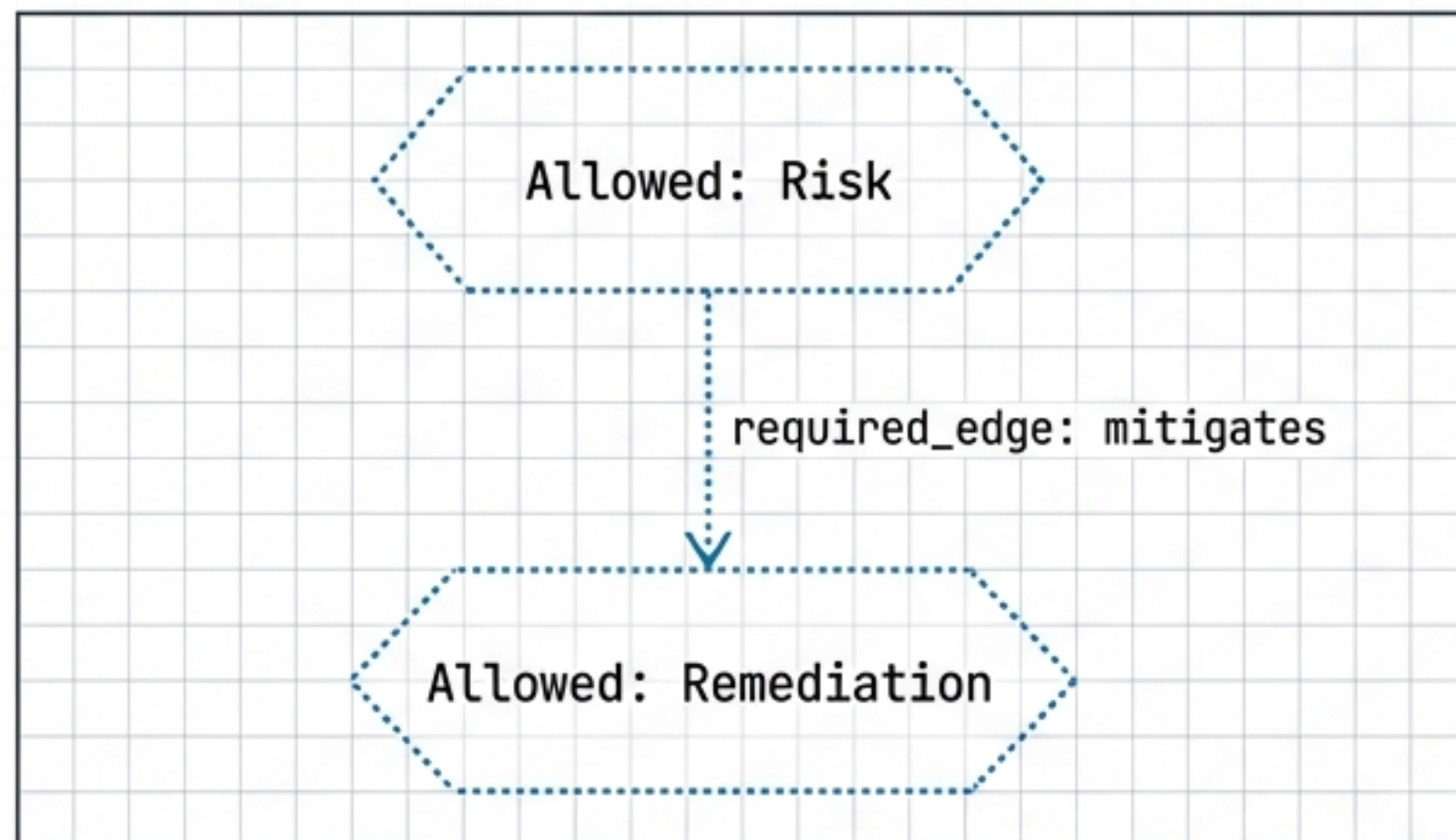
Reduction (Zoom Out) - Summary & Projection

Expansion (Zoom In) - Detail & Context

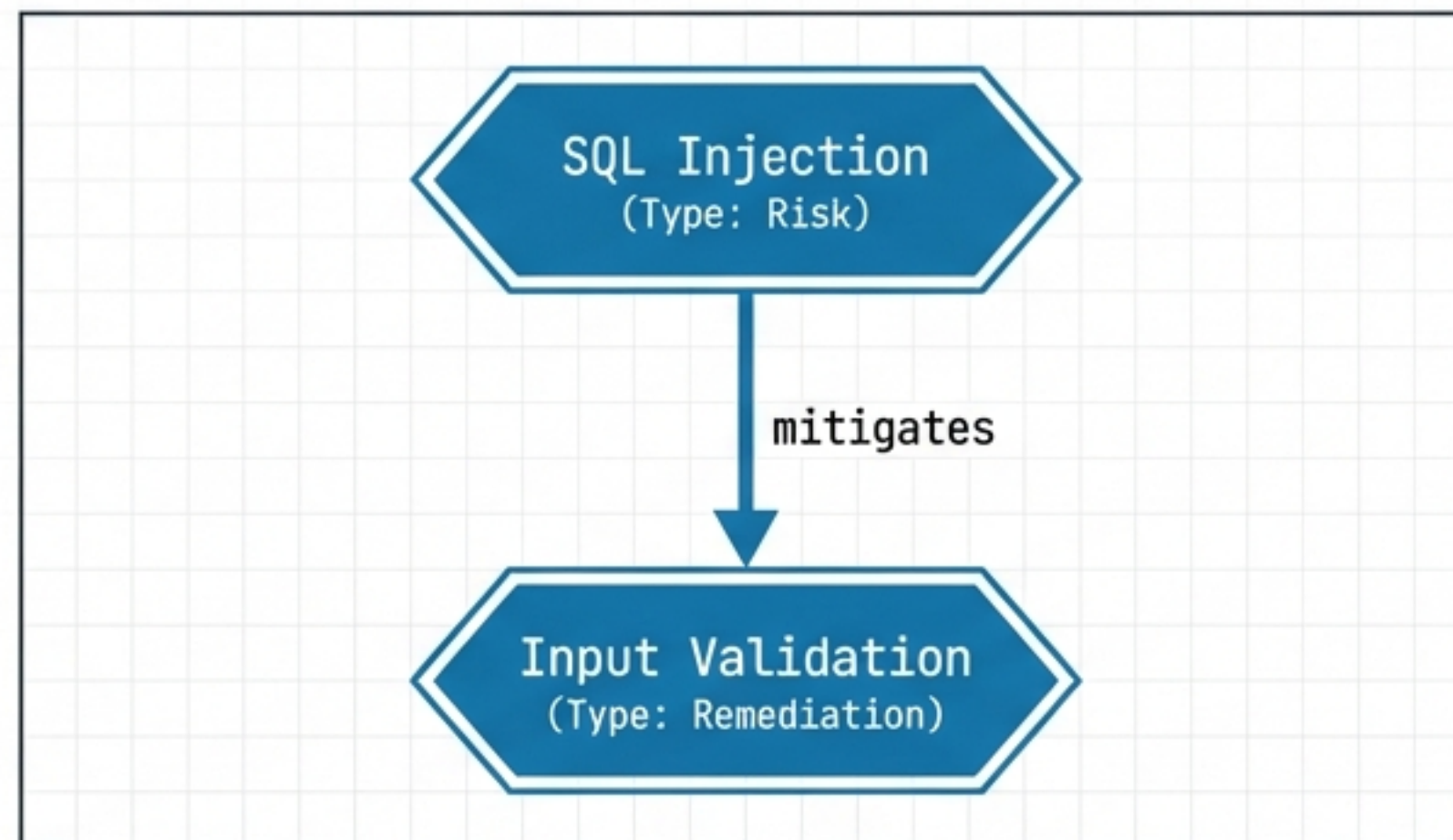
O&T First: The Contract for Extraction

Defining the rules before processing prevents graph sprawl.

The Blueprint (O&T)



The Extraction (Result)



Drift Detection Mechanisms



Structural Drift (Missing Node)



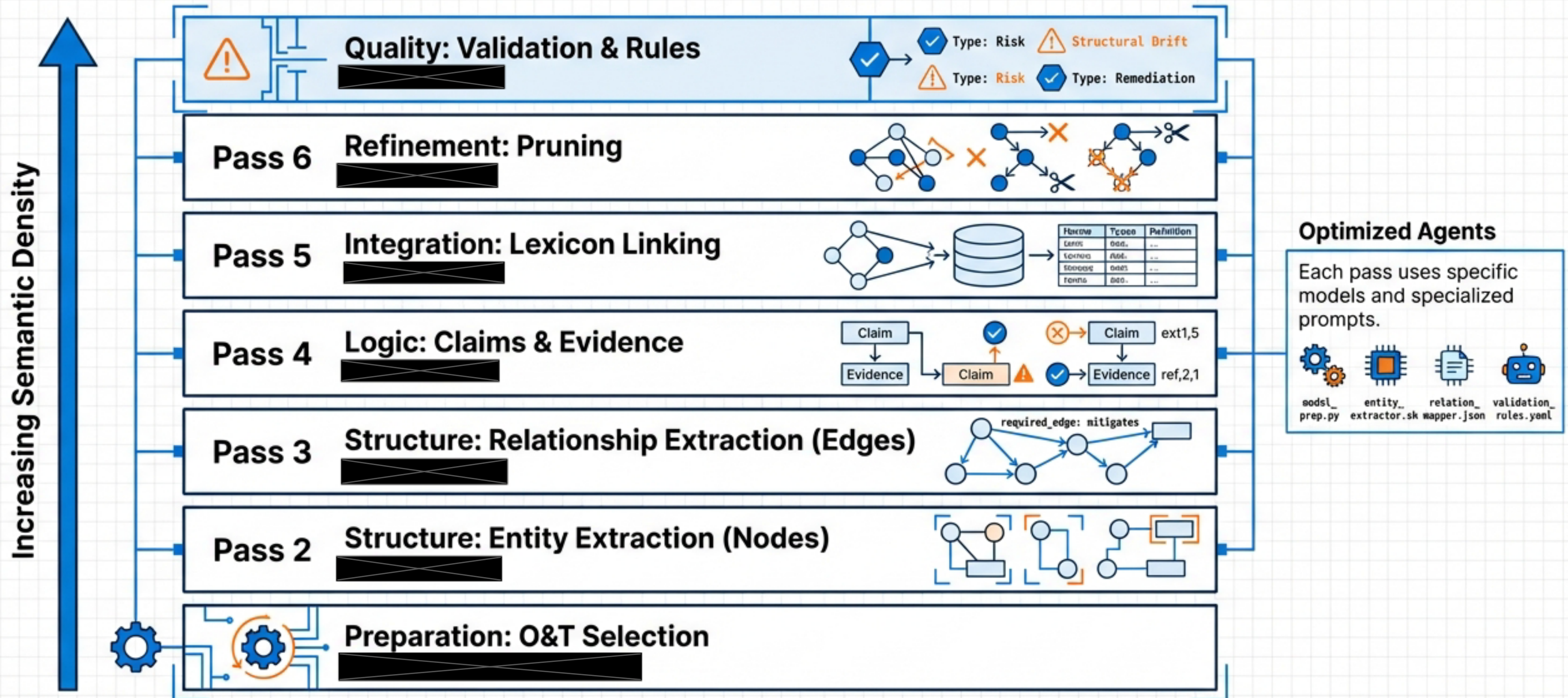
Vocabulary Drift (Wrong Term)



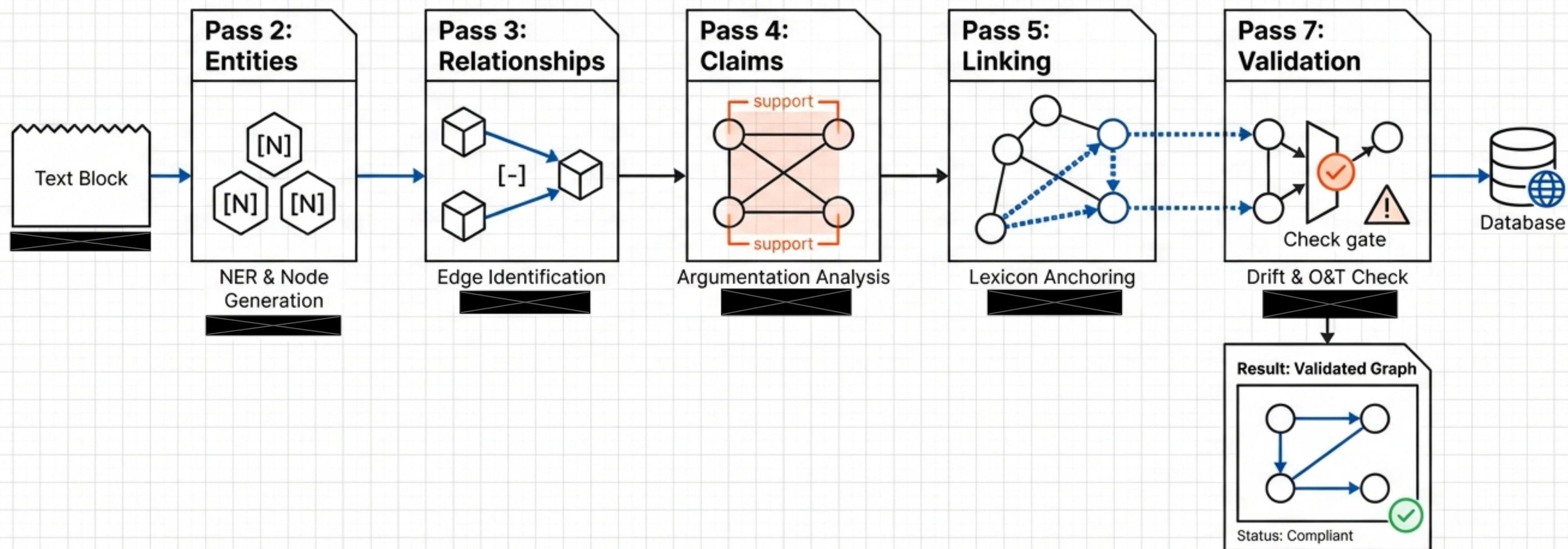
Scope Drift (Unexpected Concept)

The Multi-Pass Extraction Strategy

Moving from single-prompt chaos to separation of concerns.



Deep Dive: The Extraction Pipeline



Link Dynamics: Static vs. Auto-Generated

Static Links



Origin: Human Intent
Trust: High / Verified
Persistence: Permanent until removed

Auto-Generated Links



Origin: LLM Inference
Trust: Provisional / Low
Persistence: Re-evaluate on text change

Reactive Workflows

Source Text Deleted



Stale Link Detection



Orphaned Auto-Links Removed

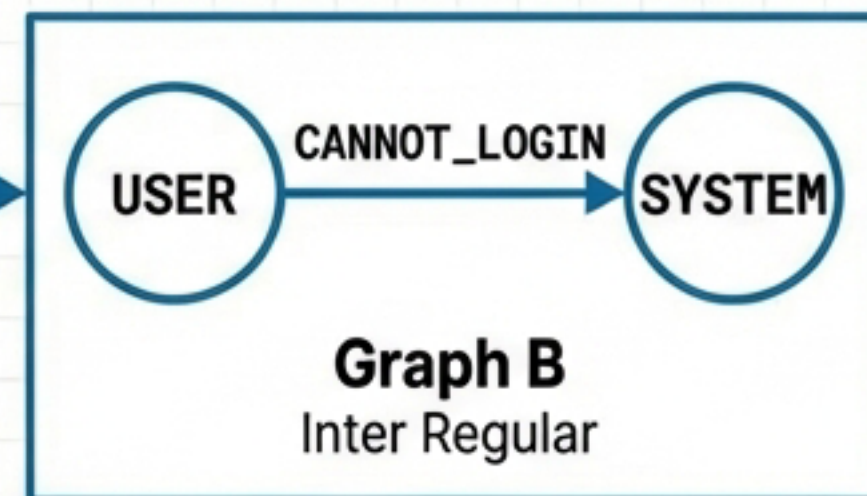
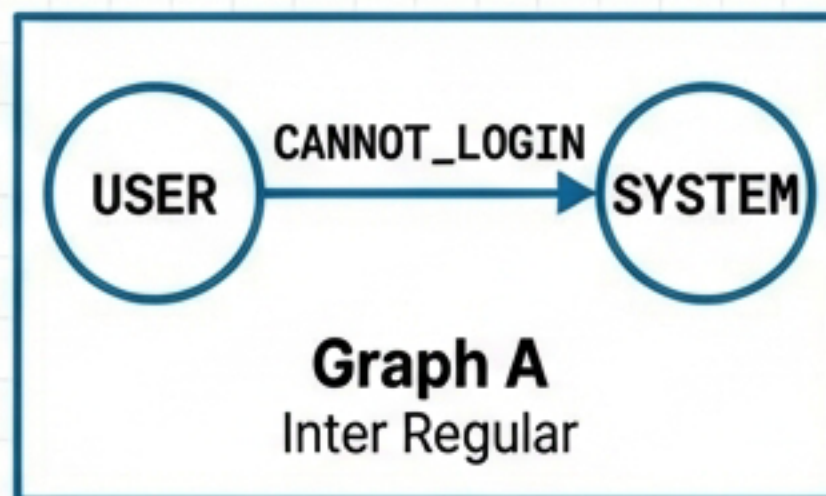


Beyond Text Diff: Detecting Semantic Change

Scenario: Rewording (No Semantic Change)

Text A: "User cannot login."

Text B: "The user is unable to log in."



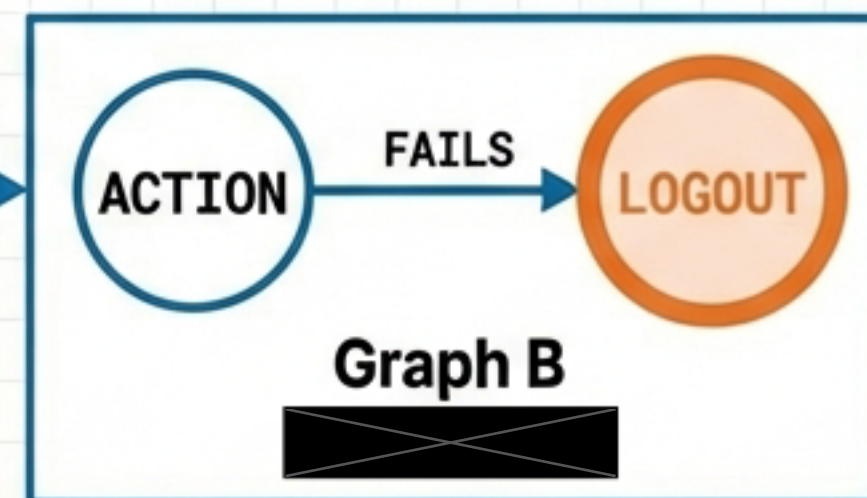
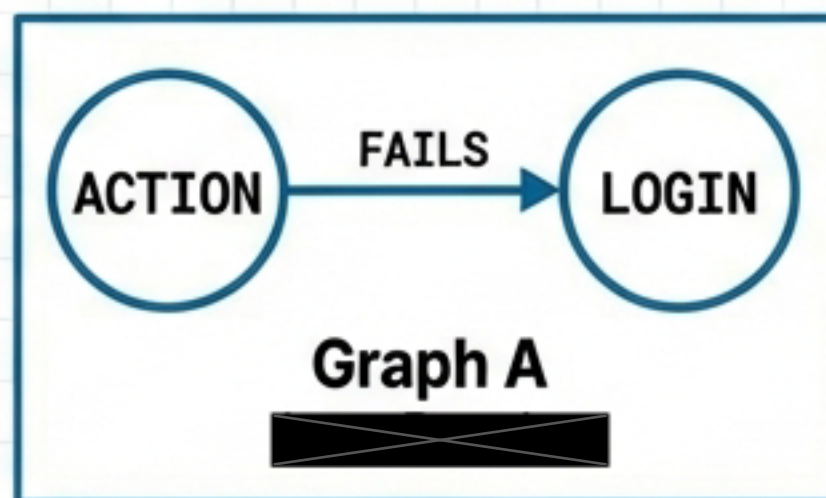
Graph Diff: 0 changes. No Action Required.



Scenario: Meaning Change

Text A: "Login fails."

Text B: "Logout fails."



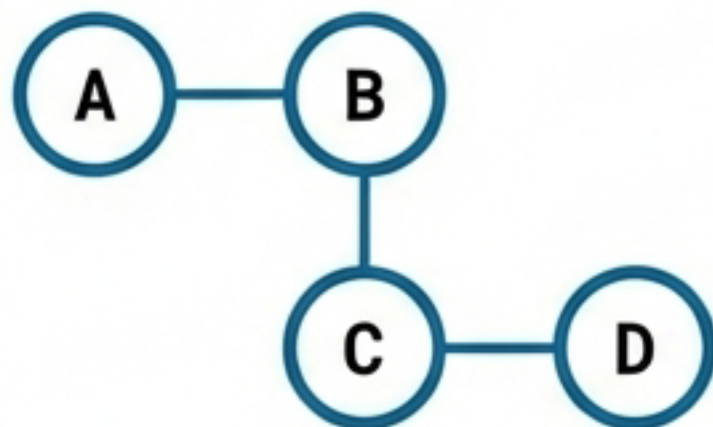
Graph Diff: Node Mismatch. **Action:** TRIGGER REVIEW.



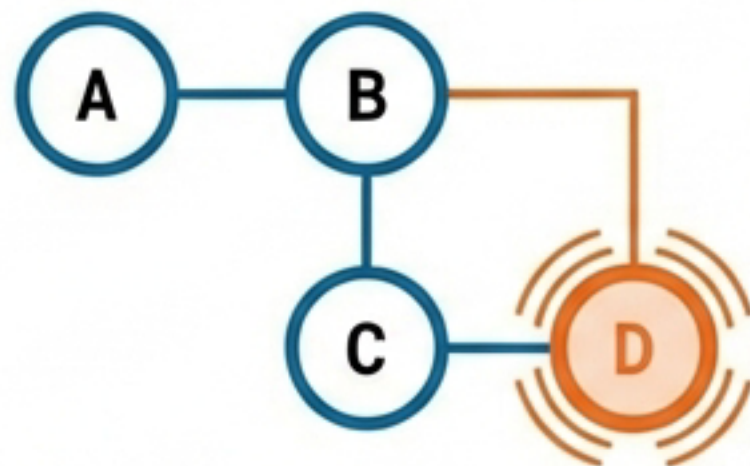
Safeguards: Hallucination & Drift Detection

Hallucination Check

Source Graph



Summary Graph



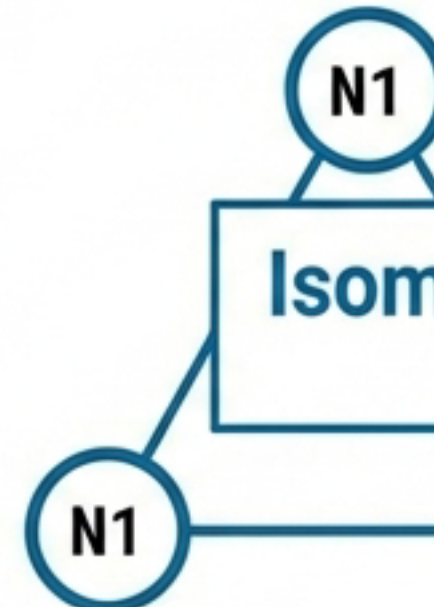
Summary contains Node D.
Source does not contain Node D.

Result: **HALLUCINATION DETECTED**

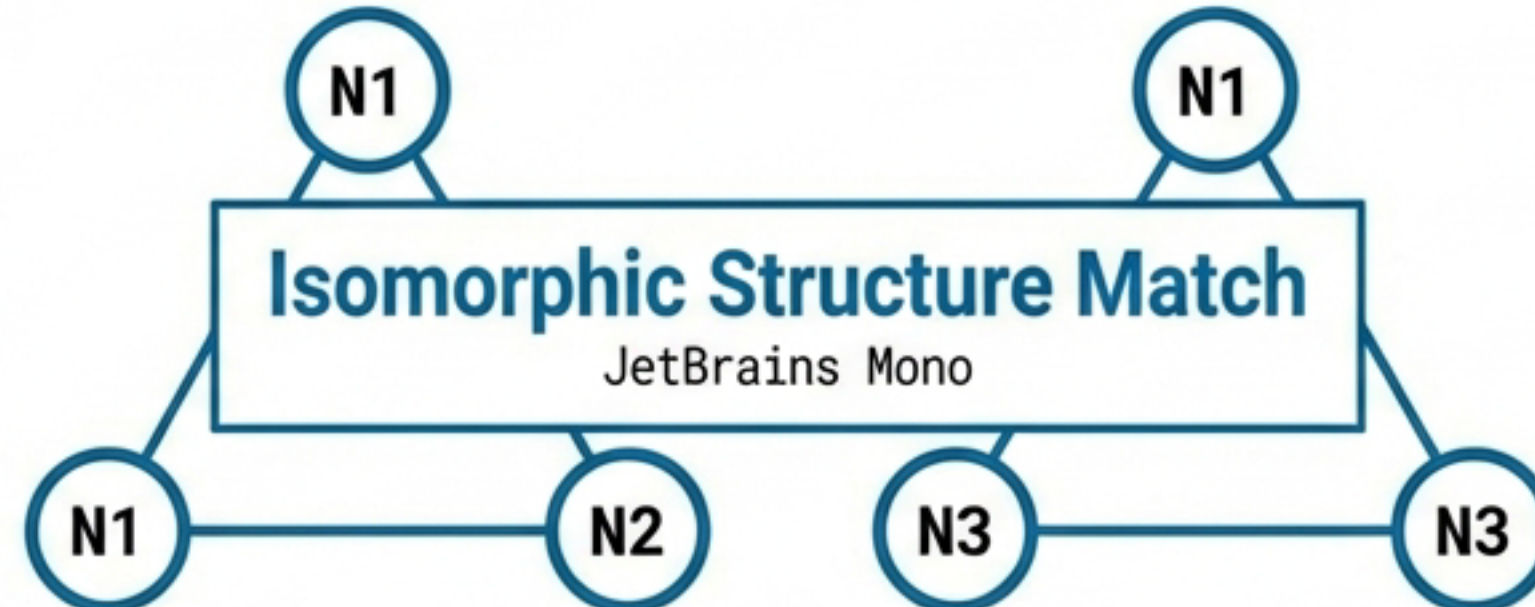


Translation Validation

English Graph



Japanese Graph

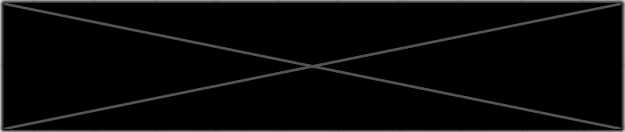


Result: **VALID TRANSLATION**

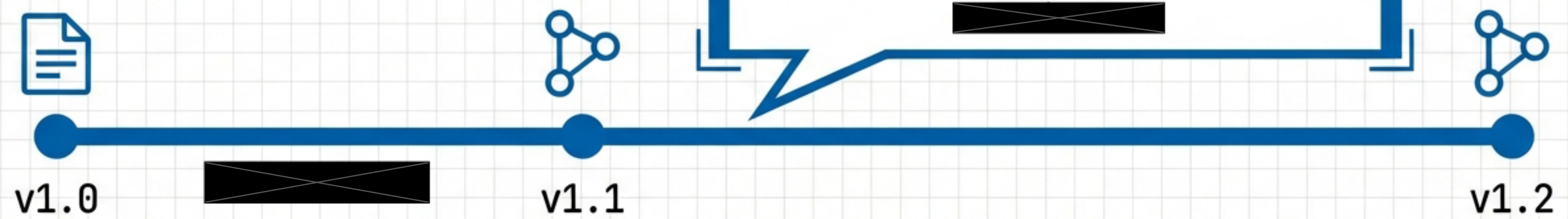


External Drift: Detects when external policy documents change, invalidating local graph edges.

Semantic Version Control

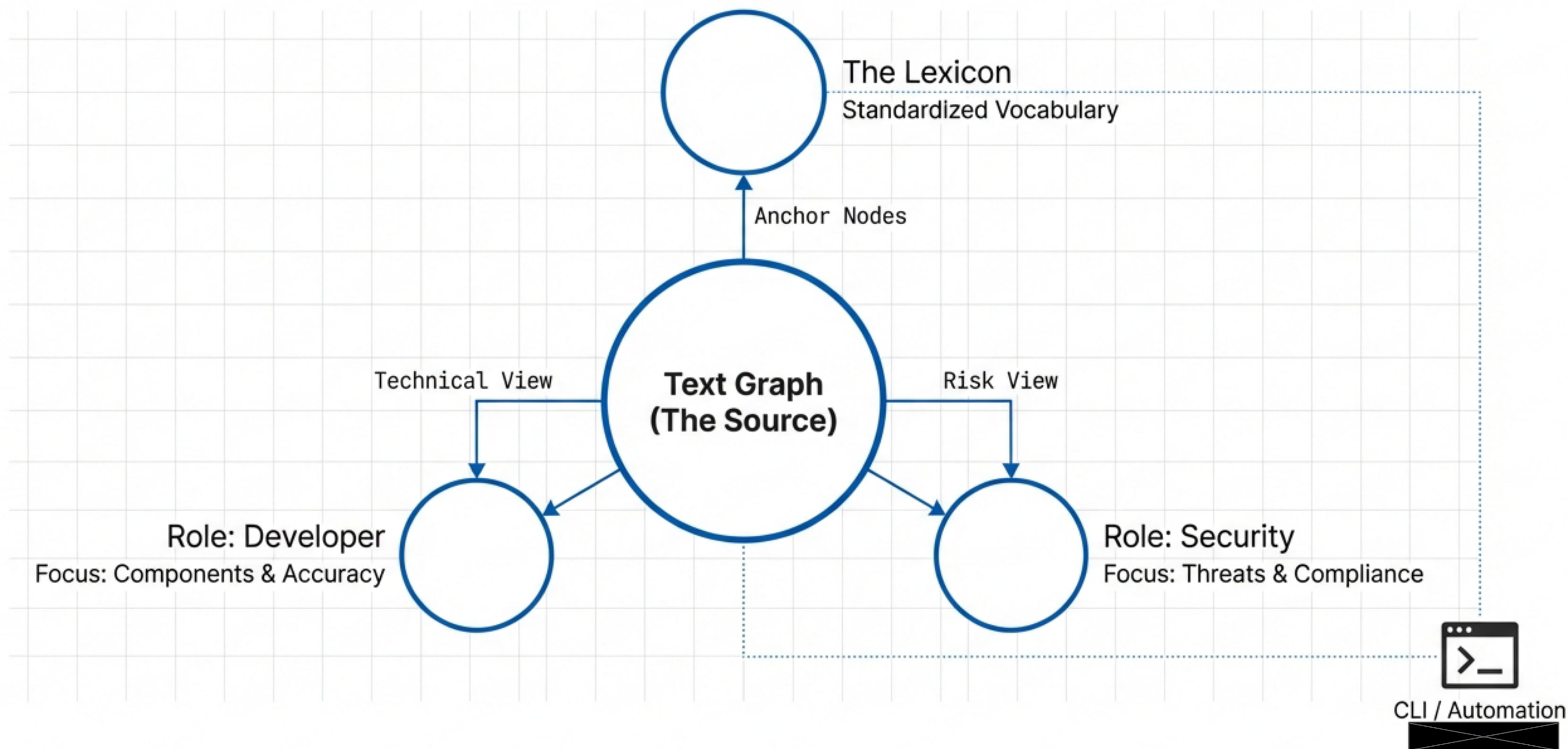
Git for Meaning. 

Semantic Blame: User 'jdoe' changed 'Requirement_A' logic.

Action Triggers 		
Trigger Event 		Automated Action 
Minor Text Edit		Bypass Regression Test ✓
Semantic Logic Change		Enforce Strict Review ⚠

Ecosystem Integration: Lexicon & Roles



Strategic Decisions Log

ID	Decision	Rationale
ST1	Text is a graph, not a string	Enable non-linear analysis & validation.
ST2	O&T first, extraction second	Prevent graph sprawl; enable drift detection.
ST3	Multi-pass extraction	Separation of concerns for higher quality.
ST4	Static vs auto-generated links	Trust assessment and stale link detection.
ST5	Independent Database Levels	Fractal storage, loading, and versioning.
ST6	Graph diff, not text diff	Trigger actions based on meaning, not syntax.
ST7	Concepts must link to Lexicon	Verified meaning across the ecosystem.
ST8	Version history as graphs	Semantic Git operations (blame/diff).

The Vision: Testable, Transparent Content

**Text is no longer the end of
of the graph.**

It is just the beginning.

By treating text as a graph, we move from passive documentation to active, intelligent knowledge bases that can be tested, validated, and queried like code.

References

- "Thinking in Graphs: Meaning Through Connectivity"
- "Issues-FS Lexicon Architecture"
- "Issues-FS Architecture Overview"